

30번

In humans, the infant immune system is less active than that of adults, enabling a wide range of bacteria to establish in our guts. Similarly, young plants release fewer defensive compounds into the soil than older ones, allowing a broad variety of microbes to colonize their rhizospheres. Human breast milk contains sugars. At first, scientists struggled to understand why mothers express these compounds, as babies can't digest them. It now seems that their sole purpose is to feed the bacteria with which the child will grow. They selectively cultivate a particular bacterial species with a crucial role in helping the gut to develop and fine-tuning the immune system. Similarly, young plants release large quantities of sugars into the soil, to feed and develop their new microbiomes. Like the human gut, the rhizosphere not only digests food, but also helps to protect plants from disease. Just as the bacteria that live in our guts outcompete and attack invading pathogens, the microbes in the rhizosphere create a defensive ring around the root. Plants feed beneficial bacteria species, so that they eliminate pathogenic microbes.